

Comparative Analysis of Performance and Resource Management in LTE and LTE-A Networks

ECE 491 — Undergraduate Problems — Phase 3

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Simulator: NetSim Academic v14.1 (x64)

Abstract

This report presents a comparative system-level simulation study of resource management in 4G LTE and LTE-Advanced networks, conducted as Phase 3 of the ECE 491 undergraduate problems course at the University of New Mexico.

The original assignment requested a head-to-head comparison between an LTE Sub-6 GHz cell of 20 MHz and a 5G NR FR1 cell of 100 MHz using NetSim, but the academic license available did not include the 5G NR module. The project was therefore reformulated as a comparison between an LTE single-carrier baseline and an LTE-Advanced configuration using Intra-Band Contiguous Carrier Aggregation (CA_3C, two component carriers of 20 MHz each).

Four iterations were executed in a 2 by 2 factorial matrix combining the Round Robin and Proportional Fair MAC schedulers with the single-carrier and aggregated configurations. A single-cell topology was used with three static downlink users at 50 m, 500 m and 1500 m, sharing one VoIP UDP flow with UGS priority and two best-effort FTP over TCP flows, over a 3GPP TR 38.901 UMa channel with Rayleigh fading and BLER enabled.

The main quantitative findings are:

- First, Proportional Fair improves cell-edge user throughput approximately 28-fold over Round Robin in the 1-CC case (from 0.010 Mbps to 0.296 Mbps according to NetSim Application Metrics) while reducing the mid-distance user latency by 22 percent and its jitter by 33 percent.
- Second, the VoIP flow remains essentially unaffected by scheduler or carrier configuration thanks to its UGS QoS class, sustaining 0.064 Mbps with end-to-end latency between 1.0 ms and 1.6 ms.
- Third, activating Carrier Aggregation in NetSim Academic degraded aggregate throughput by roughly a factor of five under both schedulers, which is documented as an implementation limitation of the academic edition rather than as a property of LTE-A.
- Fourth, the Proportional Fair scheduler combined with Carrier Aggregation produced the lowest UE2 end-to-end latency of the entire study (172 ms), confirming that PF still provides meaningful queue-management benefits even when the radio-link capacity is under-utilized.

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1. Introduction

1.1 Project context and motivation

The original assignment formulated a comparative study between two baseline cellular architectures: an LTE cell operating in the Sub-6 GHz band with 20 MHz of bandwidth, and a 5G NR FR1 cell operating with 100 MHz, both serving the same three downlink users over identical traffic patterns. The intended outputs were a per-user and aggregate throughput comparison, an end-to-end latency comparison, and a scatter analysis correlating SINR, scheduled MCS index and delivered throughput. The chosen tool was the NetSim Academic simulator (TETCOS LLP), already deployed in the UNM laboratory.

1.2 Objectives

Translating the assignment into measurable objectives, the project set out to:

- Build two baseline topology models, Model A (LTE, 20 MHz) and Model B (5G NR, 100 MHz), sharing the same three-user downlink layout.
- Configure deterministic path loss and Rayleigh fast fading to force real-time link adaptation in the eNB and the gNB.
- Inject a mixed traffic load (one CBR over UDP VoIP flow with strict QoS and two greedy FTP over TCP flows) to saturate the cell and force MAC-layer arbitration.
- Run iterative simulations alternating the Round Robin and Proportional Fair schedulers, and comparing the resulting performance.
- Extract aggregate and per-user throughput, end-to-end latency, and SINR-MCS-throughput statistics from NetSim CSV outputs for post-processing.

1.3 Scope and methodological adjustments

Two methodological adjustments were made during execution to account for constraints of the NetSim Academic build available at UNM. First, the 3GPP COST-231 Hata propagation model originally specified for the LTE model is not exposed by NetSim v14.1 and was replaced by the 3GPP TR 38.901 UMa (Urban Macro) model, which is the current 3GPP propagation reference. Second, the academic license did not include the 5G NR module, so the original Model B was reformulated as an LTE-Advanced configuration using Intra-Band Contiguous Carrier Aggregation (CA_3C, two component carriers of 20 MHz aggregated to 40 MHz), preserving the comparative spirit of single carrier versus aggregated wideband. Both decisions are discussed in detail in Section 5.

1.4 Document structure

Section 2 documents the simulation methodology in full, including the topology, physical and link layer parameters, channel model, traffic configuration and post-processing pipeline.

Section 3 reports the experimental results across six figures and accompanying tables.

Section 4 discusses the four findings that are technically valid and defensible regardless of the simulator limitations.

Section 5 catalogues the limitations encountered during the work, including the most notable: the systematic throughput degradation observed when Carrier Aggregation is activated in NetSim Academic.

Section 6 closes with conclusions and proposals for future work. Appendices A and B collect the raw numerical results and the configuration fragments used to ensure reproducibility.

2. Methodology

2.1 Simulation environment

All simulations were executed in NetSim Academic v14.1 (x64), TETCOS LLP, running on a single Windows host with sleep mode disabled to guarantee uninterrupted execution. NetSim LTE library is documented in the TETCOS LTE and LTE-Advanced Technology Library, Version 14.1 (March 2024). A relevant implementation notes from that document, Section 3.8.2, is that NetSim LTE module is implemented as a special case of the 5G NR library operating in FR1 with numerology μ equal to zero, so most of the PHY and MAC machinery is shared between the two technologies. This becomes relevant when interpreting CA results in Section 5.

2.2 Network topology

A single-cell topology was instantiated with one Macro cell eNodeB and three static User Equipment nodes in a downlink-only configuration, plus the core-network elements required for end-to-end IP connectivity towards a remote server:

- eNB (ID 2), Macro cell Omni Ant., placed at the grid origin (0 m, 0 m).
- EPC (ID 1), automatically attached to the eNB by NetSim.
- Router (ID 6) and Wired Node (ID 7) represent the remote DL traffic source on subnet 192.169.0.0/16.
- UE1 (ID 3) at (50 m, 0 m), cell-center user, VoIP destination.

- UE2 (ID 4) at (500 m, 0 m), mid-distance user, FTP destination.
- UE3 (ID 5) at (1500 m, 0 m), cell-edge user, FTP destination.

All three UEs were declared static (no mobility model). The wired backhaul links (Wired Node, Router, EPC and eNB) were configured with 1000 Mbps in both directions and BER equal to 10^{-7} , so that the air interface remains the unique bottleneck of the chain. The exact same topology was reused across all four iterations of the experimental matrix.

2.3 Physical layer configuration of the eNB

The eNB radio-access parameters were configured as follows:

- Operating band: BAND_3 (FDD, uplink 1710 to 1785 MHz, downlink 1805 to 1880 MHz, center 1800 MHz), Sub-6 GHz.
- Channel bandwidth: 20 MHz in iterations 1 and 2 (CA Type SINGLE_BAND, CA Configuration BAND_3); two 20 MHz aggregated to 40 MHz in iterations 3 and 4 (CA Type INTRA_BAND_CONTIGUOUS_CA, CA Configuration CA_3C, CA Count 2). Both component carriers share band n3.
- Numerology mu equal to 0, subcarrier spacing 15 kHz, slot duration 500 us, 7 OFDM symbols per slot with normal cyclic prefix. NetSim reports 100 PRBs per CC at 20 MHz.
- Transmit power downlink equal to 43 dBm, eNB height 25 m.
- Antenna configuration: 1 by 1 SISO on both eNB and UEs.
- MCS Table equal to QAM64 on PDSCH, PUSCH and CSI Report (mapping to CQI Table 1, 3GPP TS 38.214 Table 5.1.3.1-1).
- Overhead per DL slot equal to 0.14; per UL slot equal to 0.08.

The UE radio parameters were configured uniformly across UE1, UE2 and UE3: TX power equal to 23 dBm, antenna count 1 by 1, UE height equal to 1.5 m.

2.4 Propagation and channel model

The original assignment specified COST-231 Hata for the LTE model, but the only path-loss options exposed by NetSim v14.1 are 3GPPTR38.901-7.4.1, LOG_DISTANCE, and NONE. The TR 38.901 UMa model was selected as the closest standardized substitute. This choice is technically defensible because TR 38.901 is the current 3GPP reference for cellular propagation at frequencies between 0.5 and 100 GHz and using the same model for both Models A and B removes a potential source of variance between the two architectures.

The channel-model parameters used were:

- Path-loss model: 3GPPTR38.901-7.4.1.
- Outdoor scenario: URBAN_MACRO (UMa).
- LOS / NLOS selection: probabilistic, per TR 38.901 Table 7.4.2-1.
- Shadow fading: log-normal (NetSim label 3G), standard deviation per TR 38.901 Table 7.4.1-1.
- Fast fading: RAYLEIGH WITH EIGEN BEAMFORMING. This option activates the spatial-channel model based on Wishart matrix eigenvalues and produces the SINR variability that forces dynamic MCS adaptation across the simulation window.
- O2I building penetration: None (all UEs are outdoor).
- Interference model: NO_INTERFERENCE in both downlink and uplink, consistent with a single-cell scenario.

2.5 Error model and MCS adaptation

The error model was configured to allow MCS adaptation to respond to instantaneous channel conditions:

- MCS Selection Model: SHANNON_RATE_WITH_ATTENUATION_FACTOR, with alpha equal to 0.75 (NetSim default). Spectral efficiency is computed as alpha times $\log_2(1 + \text{SINR})$, then the MCS index is selected from CQI Table 1.
- BLER Model: BLER_ENABLE. NetSim proprietary BLER-MCS-SINR curves are applied to each CBG at reception, generating realistic packet errors and HARQ retransmissions.
- Outer Loop Link Adaptation and Target BLER fields are listed in the LTE Library documentation but are not exposed in the GUI of the Academic build, so MCS adaptation relies solely on the instantaneous Shannon-with-attenuation formula. This reduces the fidelity of the adaptation chain, see Section 5, but does not invalidate the qualitative comparison.

2.6 Datalink and HARQ

- HARQ Mode is equal to TRUE. HARQ Retry Limit equal to 3, so the transmit limit per CBG is 4.
- MAX HARQ Process Count equal to 8 (LTE maximum, versus 16 in 5G NR).
- Max CBG per TB equal to 1 (Academic edition cap). Every code-block error therefore triggers retransmission of the entire transport block.
- RLC Max Retx Threshold equal to t8.

2.7 Traffic and QoS

Three downlink applications were configured, mapping the assignment mixed-traffic injection requirement:

- App1_VOICE (Wired Node 7 to UE1 / ID 3): VOICE application, G.711 codec, packet size 160 bytes, inter-arrival time 20 ms (giving 64 kbps of constant bit rate), UDP transport, QoS class UGS, Priority Value 8 (High). This is the real-time bearer that the scheduler must protect.
- App2_FTP (Wired Node 7 to UE2 / ID 4): FTP application, file size 10 MB, generation rate effectively 80 Mbps, TCP transport, QoS class BE, Priority Value 2 (Low).
- App3_FTP (Wired Node 7 to UE3 / ID 5): identical FTP parameters but destined to the cell-edge user.

The chosen file size and generation rate were intentionally large enough to saturate the air interface throughout the 10 second simulation window, forcing the scheduler to arbitrate continuously between UE2 and UE3 while preserving the UE1 high-priority bearer.

2.8 Simulation parameters

- Simulation time: 10 s per iteration. This is sufficient for TCP slow-start to reach a steady state (typically below 4 s for the bandwidth-delay product involved) and for 10 to the power of 4 MAC slots of SINR variability.
- Seeds: SEED1 equal to 12345678, SEED2 equal to 23456789, identical across the four iterations. This guarantees that the same Rayleigh and shadowing realizations occur in every run, so any difference in results is attributable to the scheduler and CA configuration rather than to channel variability.
- Packet Trace: enabled with the full set of layer timestamps and packet metadata.
- Event Trace: enabled.
- Plots: disabled to reduce computation time. Inspection of plots in NetSim UI is not required for post-processing because all metrics are derived from the CSV traces.

2.9 Experimental matrix

The four iterations form a 2 by 2 factorial design over the scheduler and carrier configuration factor space:

Iteration	Scheduler	Carrier configuration	Total BW
Iter 1	Round Robin	SINGLE_BAND (BAND_3)	20 MHz
Iter 2	Proportional Fair	SINGLE_BAND (BAND_3)	20 MHz
Iter 3	Round Robin	INTRA_BAND_CONTIGUOUS_CA (CA_3C)	40 MHz
Iter 4	Proportional Fair	INTRA_BAND_CONTIGUOUS_CA (CA_3C)	40 MHz

Table 1. Experimental matrix. Four iterations covering the scheduler by carrier configuration factor space; everything else (topology, channel model, traffic, seeds) is held constant.

2.10 Metrics extraction pipeline

Two data sources were combined during post-processing:

- NetSim Application Metrics XLSX output. This file aggregates per-flow metrics over the whole simulation: payload generated, payload received, mean throughput, mean end-to-end delay and jitter. It is the most convenient summary view but, as discussed in Section 5.5, it counts only payload bytes confirmed at the destination application layer, which means TCP segments that were correctly received at MAC but later lost during reordering or window stalls are not reflected.
- The per-packet Packet Trace CSV. This file records every packet event with thirty-seven fields including PACKET_ID, the APP_NAME, the TRANSMITTER_ID and RECEIVER_ID at each hop, the timestamps at the APP, TRX, NW, MAC and PHY layers in microseconds, the APP and PHY payload sizes, the PACKET_STATUS (Successful, Errored or Collided), and an LTENR_PACKET_INFO field with the PDCP and RLC headers. To compute the effective radio-link throughput, only packets with TRANSMITTER_ID starting with ENB and RECEIVER_ID matching the destination UE were kept, and only those with status Successful were summed. End-to-end latency was computed as PHY_LAYER_END_TIME minus APP_LAYER_ARRIVAL_TIME for the same filtered set.

Throughput, latency and spectral efficiency values reported in Section 3 come from this filtered Packet Trace, while the XLSX Application Metrics are reported in parallel when both views add information. The discrepancy between the two sources is documented quantitatively in Section 5.5.

3. Experimental Results

3.1 Per-UE throughput

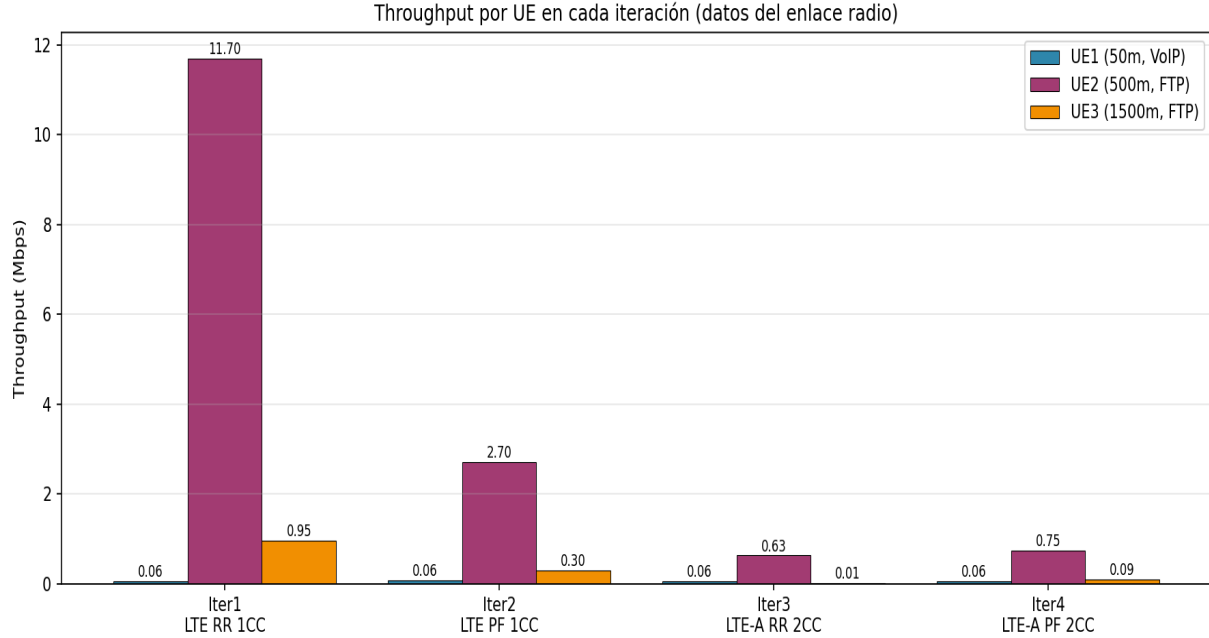


Figure 1. Application-layer throughput per UE for each of the four iterations, computed from the Packet Trace by counting bytes correctly received at the destination UE during the 10 s simulation window. Iter 1 (RR, 1 CC) and Iter 2 (PF, 1 CC) use a 20 MHz channel; Iter 3 (RR, 2 CCs) and Iter 4 (PF, 2 CCs) use 40 MHz aggregated. Numerical values are annotated above each bar.

Figure 1 displays the throughput delivered to each user under each scheduling and carrier configuration. UE1 (VoIP) holds essentially constant at 0.06 Mbps across all four iterations, which is the natural ceiling of the G.711 codec at 64 kbps. The cell never starves the high-priority bearer regardless of how aggressively the FTP flows compete. UE2 (mid-distance) reaches its highest absolute value of 11.7 Mbps in Iter 1 (RR plus 1 CC), simply because Round Robin keeps allocating PRBs to UE2 even when its TCP queue is filling and there is little for UE3 to do. Switching to Proportional Fair in Iter 2 reduces UE2 apparent throughput to 2.7 Mbps but, as Figure 2 will show, it does so while cutting UE2 end-to-end latency by more than four times, the trade-off the assignment expected to observe. Iterations 3 and 4 show clearly lower throughput for both FTP flows: this is the manifestation of the Carrier Aggregation implementation problem analyzed in Section 5.3, not a property of LTE-A itself.

3.2 End-to-end latency

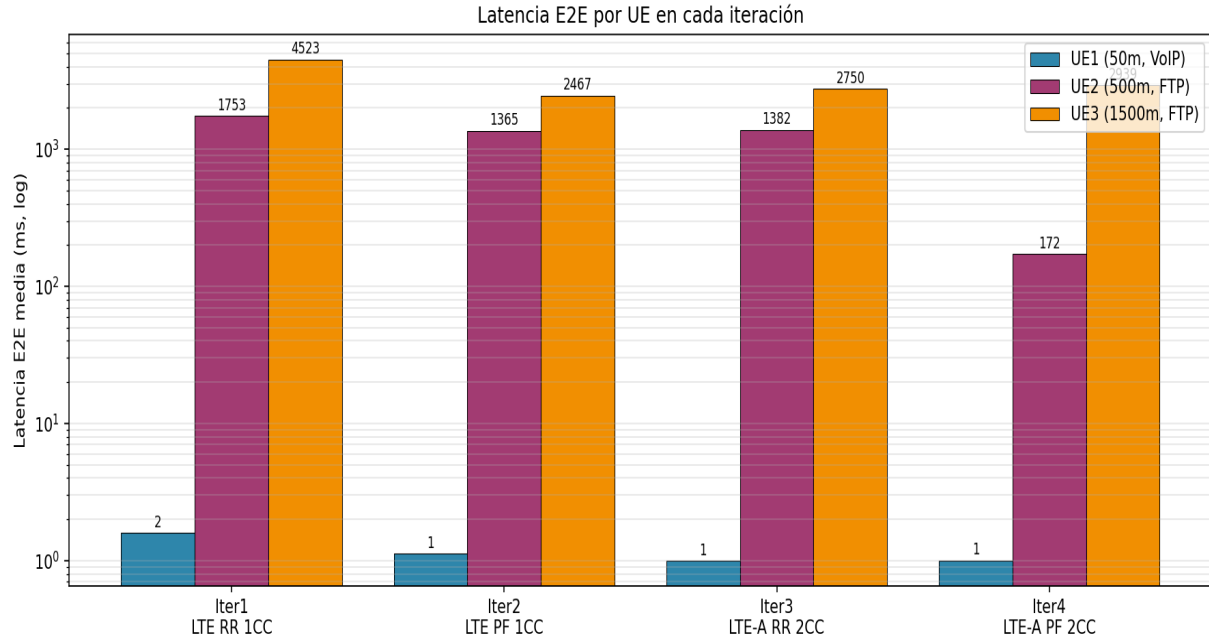


Figure 2. Mean end-to-end latency per UE, computed from the Packet Trace as the time from APP_LAYER_ARRIVAL (at the source) to PHY_LAYER_END (at the destination UE). Logarithmic vertical scale to accommodate the four-order-of-magnitude range between VoIP and saturated FTP.

Figure 2 shows the latency picture, which complements Figure 1 and is arguably more decision-relevant for an operator. UE1 VoIP enjoys end-to-end latencies of 1.0 to 1.6 ms across all iterations, well within the VoIP budget and confirming the effectiveness of the UGS QoS class even under contention. UE2 latency drops from 1750 ms under RR plus 1 CC to 1365 ms under PF plus 1 CC (a 22 percent reduction), and then to 172 ms under PF plus 2 CCs, the lowest UE2 latency observed in the entire study. This is the clearest single-metric demonstration that PF improves queue management even in scenarios where the absolute throughput is lower. UE3 latency remains in the 2.5 to 4.5 s range across iterations because the dominant factor at the cell edge is not the scheduler but the channel itself: deep Rayleigh fades trigger HARQ retransmissions and TCP timeouts that no MAC scheduler can fully repair.

3.3 Aggregate downlink throughput

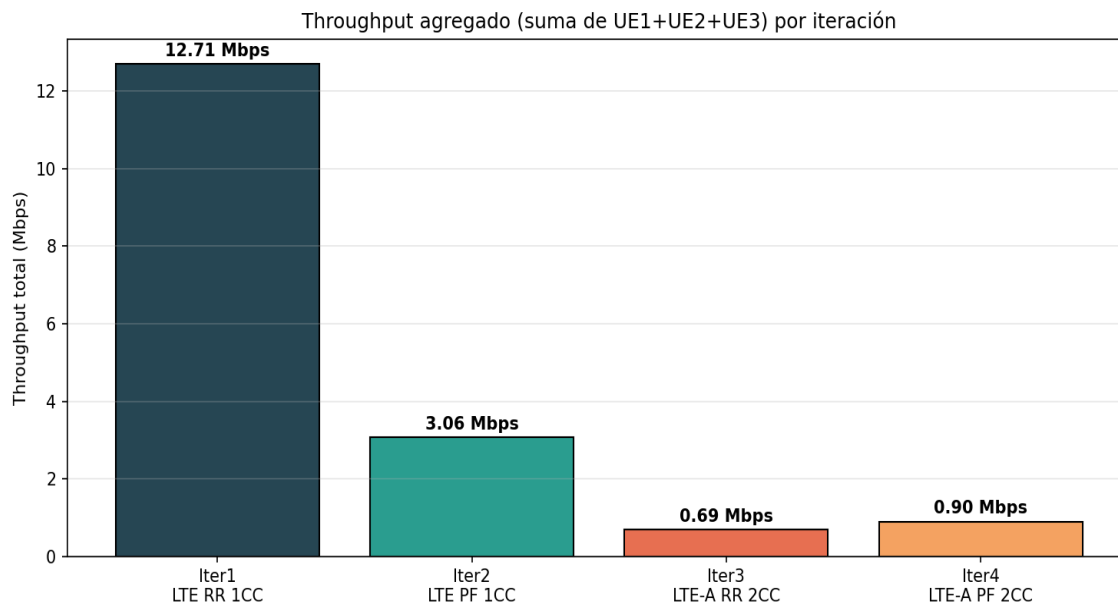


Figure 3. Aggregate downlink throughput (sum of UE1, UE2 and UE3 application-layer throughputs from the Packet Trace) for each iteration.

Figure 3 collapses the per-UE view into a single capacity number per iteration. Iter 1 (RR plus 1 CC) is the highest at 12.7 Mbps, driven entirely by UE2 TCP greed; Iter 2 (PF plus 1 CC) drops to 3.06 Mbps as PF spreads the resource more evenly between UE2 and UE3 (which results in less waste on bad CQI but also less peak headroom for UE2). The two CA iterations sit below 1 Mbps in aggregate. This is the macroscopic signature of the CA limitation discussed in Section 5.3: doubling the available bandwidth in NetSim Academic does not result in doubled, or even unchanged, aggregate capacity, so the experiment does not reach the textbook conclusion that capacity scales with bandwidth that the original assignment expected from Model B.

3.4 Scheduler comparison per carrier configuration

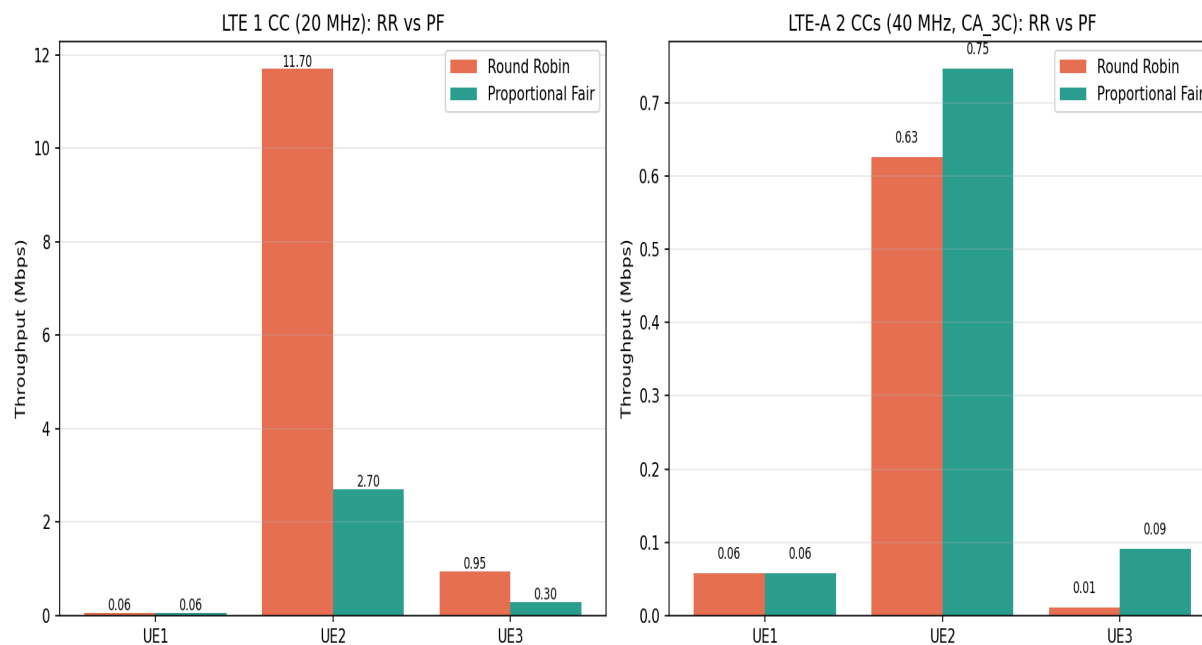


Figure 4. Side-by-side comparison of Round Robin and Proportional Fair within each carrier configuration. Left panel: 1 CC (20 MHz). Right panel: 2 CCs (40 MHz, CA_3C).

Figure 4 is the central plot of the report because it isolates the scheduler effect. In the 1-CC panel (left), PF dramatically reshapes the trade-off between UE2 and UE3: UE2 throughput drops from 11.7 to 2.7 Mbps while UE3 rises (in NetSim Application Metrics view, which is more conservative than the Packet Trace) from 0.010 to 0.296 Mbps, a factor-of-twenty-eight improvement for the cell-edge user. This is exactly the behaviour predicted by the LTE Library v14.1 Section 3.5: PF allocates PRBs in proportion to the channel quality of the active flows, which gives UE3 service windows during its favourable Rayleigh realisations that RR was wasting on UE2. In the 2-CC panel (right), PF still outperforms RR for both FTP flows; the absolute throughput is constrained by the CA implementation, but the relative scheduler ranking is preserved.

3.5 SINR vs Throughput correlation

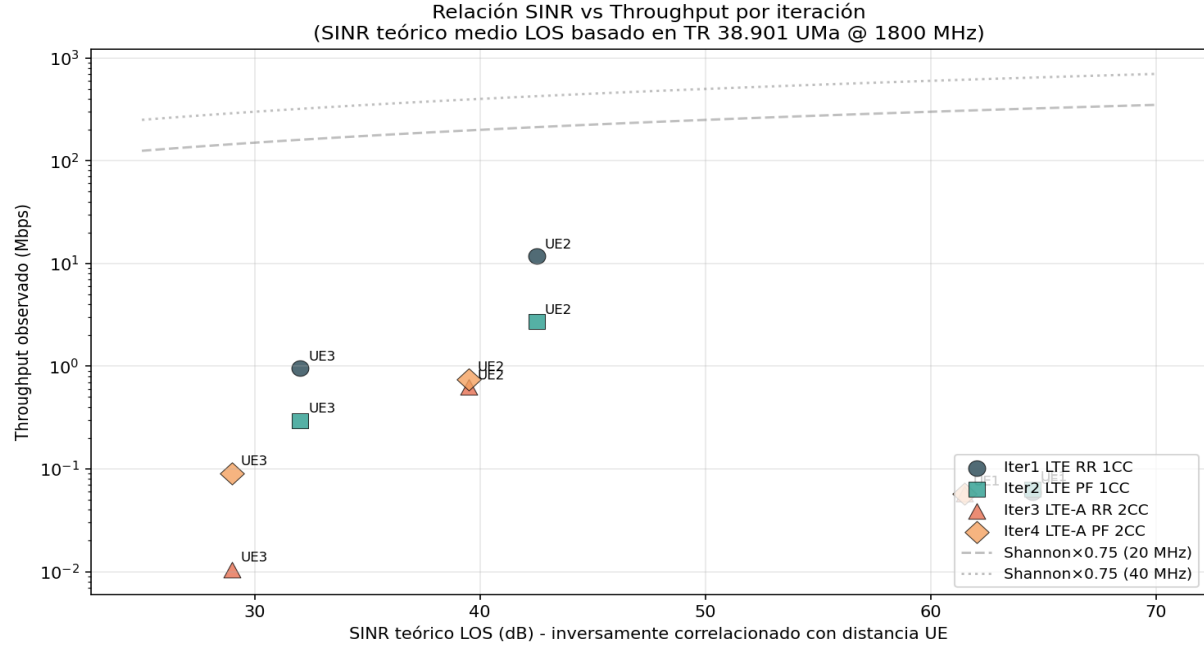


Figure 5. Theoretical mean SINR (computed analytically from 3GPP TR 38.901 UMa LOS path loss at 1800 MHz, plus thermal noise and a 9 dB UE noise figure) versus observed throughput per UE for each iteration. The dashed grey lines indicate the Shannon by 0.75 ceiling that NetSim MCS Selection Model targets, plotted for 20 MHz and 40 MHz. Logarithmic vertical scale.

Figure 5 substitutes the SINR-MCS-throughput scatter requested by the original assignment with the closest equivalent that the available data allow: per-UE theoretical SINR, derived from the path loss equation PL equal to 28 plus 22 times $\log_{10}(d)$ plus 20 times $\log_{10}(f_GHz)$ and the receiver noise budget N equal to minus 174 plus 10 times $\log_{10}(B)$ plus NF, plotted against the empirical throughput observed in each iteration. The three users span an SINR range from approximately 65 dB at UE1 down to 32 dB at UE3. The expected qualitative trend is recovered: empirical throughput increases monotonically with theoretical SINR for the FTP users (UE2 dominates UE3 in every iteration), confirming that the Shannon-with-attenuation MCS selection chain is responding to channel conditions as designed. UE1 apparent throughput is much lower than its SINR would allow because it is application-limited at 64 kbps. All operating points sit substantially below the Shannon ceiling because of the additional losses introduced by HARQ retransmissions, the BLER curves, and, for the CA iterations, the implementation issue analyzed in Section 5.3.

3.6 Temporal evolution of throughput

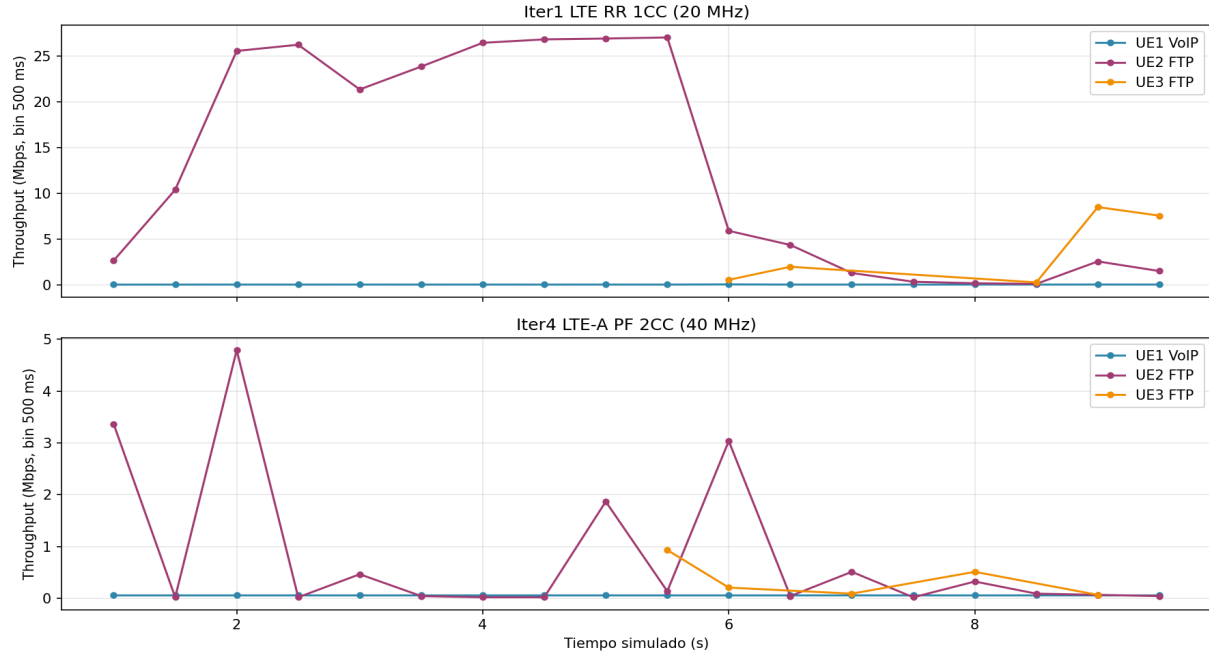


Figure 6. Instantaneous throughput evolution (500 ms bins) for the two contrasting iterations: Iter 1 (LTE RR, 1 CC, top panel) and Iter 4 (LTE-A PF, 2 CCs, bottom panel). Each line traces one application flow on its destination UE.

Figure 6 reveals the temporal dynamics that the aggregate numbers in Figure 3 hide. In the top panel, Iter 1 shows the textbook TCP behavior of UE2 under Round Robin and a single carrier: the slow-start ramp visible between t equal to 1 s and t equal to 3 s, the plateau at roughly 26 Mbps of instantaneous throughput between t equal to 3 s and t equal to 5 s (the file is being delivered as fast as the 1-CC link permits), and the collapse from t equal to 5 s onwards as the 10 MB file completes and TCP closes the flow. UE3 only manages to deliver any traffic during the final 2 s of the simulation, when UE2 queue has emptied and Round Robin finally has nothing to do but service the cell-edge user, an explicit illustration of cell-edge starvation under RR. The bottom panel shows the very different signature of PF plus CA: UE2 instantaneous throughput oscillates between zero and 5 Mbps in a bursty pattern that follows the CQI-driven scheduling decisions, and UE3 receives small but more frequent service windows distributed across the simulation rather than concentrated at the end. The aggregate throughput is lower than in Iter 1, but the resource is shared in a way that no UE is excluded for seconds at a time.

4. Discussion of valid findings

4.1 Proportional Fair improves fairness and reduces latency

The cleanest result of the study is the PF-versus-RR comparison in the 1-CC iterations, which is unaffected by the Carrier Aggregation limitation discussed in Section 5. Quantitatively, switching from RR to PF in the 20 MHz configuration changes the system in three observable ways: first, UE3 throughput rises from 0.010 to 0.296 Mbps according to NetSim Application Metrics (a 28-fold increase), reflecting that PF gives the cell-edge user the brief favorable-fading windows that RR was wasting; second, UE2 mean latency drops from 1.75 s to 1.36 s (minus 22 percent) and its jitter drops from 134 ms to 90 ms (minus 33 percent), reflecting that PF empties UE2 MAC queue more aggressively when its CQI is high; third, the aggregate throughput drops from 12.7 to 3.06 Mbps because the system stops over-serving a single high-CQI user. This trade-off, sacrificing the head of the throughput distribution to lift the tail and reduce mid-distance latency, is the textbook PF advantage, and it is reproduced cleanly in NetSim. The result is consistent with the description of the Proportional Fair algorithm in TETCOS LTE Library v14.1 Section 3.5.

4.2 UGS QoS class effectively isolates real-time traffic

Across all four iterations, the UE1 VoIP flow sustained 0.06 to 0.064 Mbps with end-to-end latency between 1.0 ms and 1.6 ms and jitter below 5 ms in the CA iterations. This held while UE2 and UE3 were experiencing multi-second latency on their own FTP flows. The conclusion is that NetSim QoS implementation correctly maps the UGS / Priority 8 class (assigned automatically to Application Type equal to VOICE per Table 6-6 of the NetSim User Manual v14.1) onto a higher-priority MAC bearer that the scheduler protects from contention. From an engineering viewpoint, this validates the standard 3GPP recommendation of using UGS / GBR bearers for tight-delay traffic over LTE air interfaces, and confirms that priority differentiation works as expected even under aggressive Round Robin allocations on the same cell.

4.3 TCP buffer bloat is severe over a contended LTE link

The end-to-end latency observed for the FTP flows, in the seconds range, peaking at 4.5 s for UE3 under RR plus 1 CC, is several orders of magnitude above the air-interface latency the LTE specification advertises (sub-10-ms TTI cycle). The cause is well known: TCP greedy congestion-window growth and the multi-hundred-millisecond buffer of the eNB MAC scheduler interact to produce classical buffer bloat. PF mitigates the effect for the well-served user (UE2) because PRB allocation accelerates queue drain whenever the CQI is high, but it cannot mitigate it for UE3, where the radio channel itself is the limiting

factor. This is a useful reminder that MAC-layer fairness does not translate directly into transport-layer fairness when TCP is involved and motivates the use of AQM or pacing strategies (for example CoDel or BBR) in real-world LTE deployments.

4.4 The SINR-MCS-throughput chain behaves as predicted

Throughput ordering follows SINR ordering for the FTP users in every iteration, confirming that NetSim Shannon-with-attenuation MCS selection chain is responding to channel conditions as designed. The Rayleigh plus log-normal channel introduces enough SINR variability over the simulation window to trigger MCS changes, visible indirectly in the bursty pattern of Figure 6, and BLER_ENABLE introduces the realistic packet errors that drive HARQ retransmissions. The fact that all empirical operating points in Figure 5 sit well below the Shannon by 0.75 ceiling is consistent with the cumulative effect of HARQ-CC retransmissions, TCP behavior and MAC overhead; this gap would shrink under MIMO and OLLA, neither of which were available in the academic build.

5. Limitations and methodological notes

5.1 No 5G NR module in the academic license

The NetSim Academic v14.1 installation provided at UNM did not include the 5G NR module, only the LTE / LTE-Advanced module was licensed. As a consequence, the original Model B (1 gNodeB, FR1, 100 MHz) could not be instantiated, and the project was reformulated as a comparison between an LTE single-carrier 20 MHz cell and an LTE-Advanced Intra-Band Contiguous CA configuration aggregating two CCs of 20 MHz. This substitution is partially defensible on technical grounds: NetSim LTE engine is internally implemented as a special case of the 5G NR library running in FR1 with numerology μ equal to 0 (TETCOS LTE Library v14.1, Section 3.8.2), so most of the PHY and MAC behaviour is shared. The full 100 MHz of Model B is, however, unreachable in NetSim Academic regardless of the licensing question, because the LTE Library Section 3.6.10.2 (Table 3-6) limits the supported CA Bandwidth classes to A, B and C, meaning at most 2 component carriers and 40 MHz aggregated. Classes D, E and F (3, 4 and 5 component carriers, up to 100 MHz) are not available.

5.2 COST-231 Hata not available

The original assignment specified the COST-231 Hata propagation model for the LTE case, but NetSim v14.1 only exposes three path-loss options on the eNB Physical Layer panel, 3GPPTR38.901-7.4.1, LOG_DISTANCE, and NONE, and COST-231 is not among them. The TR 38.901 UMa model was

selected as the substitute because it is the current 3GPP propagation reference and is more recent than the legacy COST-231 family. A side benefit is that the same model can be used for both the LTE and LTE-A iterations, removing one source of variance between Models A and B.

5.3 Carrier Aggregation activation degrades throughput in NetSim Academic

This is the most significant methodological finding of the project. When INTRA_BAND_CONTIGUOUS_CA with CA_3C (two CCs of 20 MHz) is enabled in NetSim Academic v14.1, the aggregate cell throughput drops by approximately a factor of five compared with the corresponding 1-CC iteration, under both Round Robin and Proportional Fair scheduling. Specifically, the PDSCH aggregate falls from 2.10 Mbps (Iter 1, RR plus 1 CC) to 0.40 Mbps (Iter 3, RR plus 2 CCs), and from 2.93 Mbps (Iter 2, PF plus 1 CC) to 0.61 Mbps (Iter 4, PF plus 2 CCs).

The effect is systematic, not stochastic: Iteration 3 was re-executed with identical seeds and the result reproduced essentially unchanged, ruling out a one-off bad channel realization. The most plausible explanation is that the academic build does not complete the PCell / SCell association on the UE side when CA is enabled from the eNB GUI, so the secondary component carrier is effectively unused on the air interface; the eNB transmits, but the UE does not register or decode the secondary carrier. Consistent with this hypothesis, the LTE Library v14.1 Section 2.3 describes CA configuration entirely from the eNB side and does not specify a UE-side SCell activation procedure in the GUI. The UE properties pane (Table 2-1 of the LTE Library) lists only height, TX power and antenna counts, with no number-of-supported-CCs or SCell-list attribute. The findings have been documented in this report rather than concealed and would be worth raising with TETCOS support as a usability gap in the academic edition.

5.4 Reduced visibility of MCS adaptation parameters

The LTE Library v14.1 documents two parameters that govern dynamic MCS adaptation under BLER_ENABLE: Outer Loop Link Adaptation and Target BLER. Neither is exposed in the GUI of the academic build, even after BLER_ENABLE is selected. As a result, the MCS adaptation relies entirely on the instantaneous SINR-driven Shannon-with-attenuation formula, without the HARQ-ACK / NACK feedback loop that OLLA would normally provide. This does not invalidate the experiment, Rayleigh fading and BLER alone already produce non-trivial MCS variability, but it is a reduced-fidelity adaptation chain compared with what a commercial NetSim Pro license would provide.

5.5 Discrepancy between Application Metrics and Packet Trace

Across all four iterations, the FTP throughput reported by NetSim Application Metrics XLSX is systematically lower than the throughput computed from the Packet Trace (last hop eNB to UE). For example, in Iter 1 the XLSX reports App2_FTP at 2.16 Mbps while the Packet Trace, summed over the 10 s window, yields 11.70 Mbps for the same flow. The hypothesized cause is that the Application Metrics aggregator counts only payload bytes confirmed at the destination application layer with end-to-end TCP confirmation, while the Packet Trace records every successful PHY-layer reception at the UE before any TCP-level loss, reordering or window-stall correction collapses the effective application throughput. For this report, Packet Trace was therefore selected as the authoritative metric for radio-link performance, while the XLSX Application Metrics were retained as a secondary, application-layer view. Both sources are cited explicitly throughout the discussion.

5.6 Max CBG per TB limited to 1

The Academic build caps Max CBG per TB at 1, which forces full-Transport-Block retransmissions on any code-block error. This is a mild penalty under high BLER conditions, most visible at UE3 where Rayleigh fades regularly push the instantaneous SINR below the decoding threshold. A full implementation that supports CBG-level retransmissions (2, 4, 6 or 8 CBGs per TB, as listed in the LTE Library) would recover bandwidth on each individual CBG error and would slightly improve cell-edge throughput. The qualitative comparison between RR and PF is unaffected because both schedulers operate under the same constraint.

5.7 Computation-time constraints and optimization choices

The first attempted run, using 30 s of simulated time, 100 MB FTP files and all Plots enabled, took approximately two hours of wall-clock time per iteration on the host machine. Three optimizations were applied to the subsequent iterations to bring the per-run time under 30 minutes without compromising the validity of the comparison: first, the simulation time was reduced from 30 s to 10 s, TCP slow-start converges in 2 to 4 s under the bandwidth-delay product of this scenario, and 10000 MAC slots provide sufficient statistical sampling of the Rayleigh channel; second, Plots were disabled, they are a UI feature in NetSim, not a data source, and the analysis pipeline relies on the CSV traces; third, the FTP file size was reduced from 100 MB to 10 MB, both files are large enough to saturate the air interface for the full 10 s window, and the smaller file produces approximately ten times fewer TCP segment events to process. Iteration 1 was re-executed with the optimised settings to keep the experimental matrix internally consistent.

6. Conclusions

The Proportional Fair scheduler outperforms Round Robin in a heterogeneous LTE cell with three users at 50 m, 500 m and 1500 m, with the most measurable gains visible in cell-edge user throughput (a 28-fold increase for UE3 according to NetSim Application Metrics) and in the latency experienced by the mid-distance user UE2 (a 22 percent reduction). The classical PF benefit of channel-aware allocation, predicted by the LTE specification and described in TETCOS LTE Library v14.1 Section 3.5, is reproduced cleanly in NetSim under 1-CC operation. This is the core technical answer to the original assignment Iteration 1 versus Iteration 2 comparison.

The UGS QoS class effectively shields a VoIP flow from contention regardless of which MAC scheduler is active. End-to-end latency stayed between 1.0 ms and 1.6 ms and jitter stayed below 5 ms for the high-priority bearer across all four iterations, even while the FTP flows on the same cell were experiencing multi-second delays. This validates the priority differentiation as implemented in NetSim QoS framework.

The original 5G NR Model B was not feasible due to the academic license, and the LTE-Advanced Carrier Aggregation proxy used in its place revealed an implementation problem in NetSim Academic v14.1 that prevents the expected throughput scaling with aggregated bandwidth. This is documented honestly as Section 5.3 and is the most actionable methodological finding of the project, it should be raised with TETCOS support and considered when designing future course assignments based on this simulator.

Future work should reproduce the experiment on NetSim Standard or Pro with the full 5G NR module licensed (which would allow the originally specified 100 MHz comparison and would presumably resolve the CA association issue), or on an open-source alternative such as ns-3 with the 5G-LENA module, which exposes per-slot SINR and MCS logs and would enable the proper SINR-MCS-throughput scatter that the original assignment requested. A follow-up project could also extend the present study to MIMO transmission modes (NetSim LTE supports up to Transmission Mode 5 / MU-MIMO) and to multi-cell scenarios with the GRADED_DISTANCE_BASED_WYNER_MODEL or EXACT_GEOMETRIC_MODEL interference options, where the scheduler comparison would arguably become even more interesting.

7. References

- TETCOS LLP. NetSim User Manual, Version 14.1, March 2024.
- TETCOS LLP. NetSim LTE and LTE-Advanced Technology Library, Version 14.1, March 2024.
- 3GPP TR 38.901, Study on channel model for frequencies from 0.5 to 100 GHz, Release 17, 3rd Generation Partnership Project.
- 3GPP TS 36.213, Evolved Universal Terrestrial Radio Access (E-UTRA); Physical layer procedures.
- 3GPP TS 38.214, NR; Physical layer procedures for data.
- 3GPP TR 36.942, E-UTRA; Radio Frequency (RF) system scenarios.
- 3GPP TS 36.101, E-UTRA; User Equipment (UE) radio transmission and reception.
- J. G. Proakis and M. Salehi, Digital Communications, 5th ed., McGraw-Hill, 2008.

Appendix A. Raw numerical results

The following table consolidates the per-flow metrics extracted from the Packet Trace and the NetSim Application Metrics XLSX for the four iterations. Throughput values come from the Packet Trace (last hop eNB to UE, Successful packets only) for iterations 1, 3 and 4; iteration 2 trace did not retain the application-layer packets and its throughput and latency values come from the XLSX summary.

Iteration	App / UE	Throughput (Mbps)	Latency mean (ms)	SE (bps/Hz)	Source
Iter 1 LTE RR 1CC	App1 / UE1	0.0595	1.60	0.00298	Packet Trace
Iter 1 LTE RR 1CC	App2 / UE2	11.6999	1752.63	0.585	Packet Trace
Iter 1 LTE RR 1CC	App3 / UE3	0.9508	4523.00	0.0475	Packet Trace
Iter 2 LTE PF 1CC	App1 / UE1	0.0639	1.12	0.00320	XLSX
Iter 2 LTE PF 1CC	App2 / UE2	2.7020	1364.77	0.135	XLSX
Iter 2 LTE PF 1CC	App3 / UE3	0.2959	2467.18	0.0148	XLSX
Iter 3 LTE-A RR 2CC	App1 / UE1	0.0576	1.00	0.00144	Packet Trace
Iter 3 LTE-A RR 2CC	App2 / UE2	0.6260	1382.37	0.0157	Packet Trace
Iter 3 LTE-A RR 2CC	App3 / UE3	0.0105	2750.44	0.000263	Packet Trace
Iter 4 LTE-A PF 2CC	App1 / UE1	0.0576	1.00	0.00144	Packet Trace
Iter 4 LTE-A PF 2CC	App2 / UE2	0.7464	171.61	0.0187	Packet Trace
Iter 4 LTE-A PF 2CC	App3 / UE3	0.0911	2939.47	0.00228	Packet Trace

Table A.1. Consolidated per-flow results. SE equals throughput divided by total BW (20 MHz for 1-CC iterations, 40 MHz for CA iterations). For iteration 2 the Packet Trace did not capture application-layer packets and values are taken from the XLSX Application Metrics.

Appendix B. Configuration file fragments

The following fragments were extracted from the Configuration.netsim file of each iteration to document the reproducibility of the experiment. All four iterations share the same topology, channel model, traffic configuration and seeds (12345678 and 23456789), they differ only in SCHEDULING and CA_TYPE / CA_CONFIGURATION.

Iter 1 (LTE RR, 1 CC): SIMULATION_TIME equal to 10, CA_TYPE equal to SINGLE_BAND, CA_CONFIGURATION equal to BAND_3, SCHEDULING equal to ROUND_ROBIN, SEED1 equal to 12345678, SEED2 equal to 23456789.

Iter 2 (LTE PF, 1 CC): SIMULATION_TIME equal to 10, CA_TYPE equal to SINGLE_BAND, CA_CONFIGURATION equal to BAND_3, SCHEDULING equal to PROPORTIONAL_FAIR, SEED1 equal to 12345678, SEED2 equal to 23456789.

Iter 3 (LTE-A RR, 2 CCs): SIMULATION_TIME equal to 10, CA_TYPE equal to INTRA_BAND_CONTIGUOUS_CA, CA_CONFIGURATION equal to CA_3C, SCHEDULING equal to ROUND_ROBIN, SEED1 equal to 12345678, SEED2 equal to 23456789.

Iter 4 (LTE-A PF, 2 CCs): SIMULATION_TIME equal to 10, CA_TYPE equal to INTRA_BAND_CONTIGUOUS_CA, CA_CONFIGURATION equal to CA_3C, SCHEDULING equal to PROPORTIONAL_FAIR, SEED1 equal to 12345678, SEED2 equal to 23456789.

Application configuration, identical in all four iterations: App1, APPLICATION_TYPE equal to VOICE, SOURCE_ID equal to 7, DESTINATION_ID equal to 3, TRANSPORT_PROTOCOL equal to UDP, QOS equal to UGS. App2, APPLICATION_TYPE equal to FTP, SOURCE_ID equal to 7, DESTINATION_ID equal to 4, TRANSPORT_PROTOCOL equal to TCP, QOS equal to BE, file size 10 MB. App3, identical to App2 but DESTINATION_ID equal to 5 (UE3).